

Prog-10: Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

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1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg10
1RV24MC075_Pragathi_h_r@dell:~$ cd DevOps
1RV24MC075_Pragathi_h_r@dell:~/DevOps$ mkdir prg10
1RV24MC075_Pragathi_h_r@dell:~/DevOps$ cd prg10
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 22.04
🔗 Using the docker driver based on existing profile
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
🔗 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔗 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ minikube status
minikube
type: Control Plane
host: Running
kubernetes: Running
apiserver: Running
kubeconfig: Configured

1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ nano deployment.yaml
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ nano service.yaml
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ minikube dashboard
🔗 Enabling dashboard ...
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🔍 Verifying dashboard health ...
🔗 Launching proxy ...
🔍 Verifying proxy health ...
^C
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl apply -f deployment.yaml
error: error parsing deployment.yaml: error converting YAML to JSON: yaml: line 20: did not find expected key
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1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl apply -f service.yaml
service/nginx-service created
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
kubia                                1/1      Running   10 (7m36s ago)    23d
nginx-deployment-6f9664446b-j42pb    1/1      Running   3 (14d ago)       21d
nginx-deployment-6f9664446b-s66kz    1/1      Running   3 (14d ago)       21d
nodejs-pod                           1/1      Running   11 (7m36s ago)    21d
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl get nodes
NAME    STATUS    ROLES    AGE    VERSION
minikube Ready    control-plane 35d    v1.34.0
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl get svc
NAME                                TYPE                CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes    ClusterIP           10.96.0.1     <none>         443/TCP          35d
kubia         NodePort            10.100.219.139 <none>         8080:31947/TCP   23d
kubia-http    LoadBalancer      10.98.0.27    <pending>      8080:31841/TCP   23d
my-app        ClusterIP           10.100.102.194 <none>         80/TCP           30d
my-nginx-pod  ClusterIP           10.99.114.205 <none>         80/TCP           30d
nginx-deployment NodePort            10.110.113.243 <none>         80:32260/TCP     21d
nginx-service NodePort            10.109.150.144 <none>         80:32401/TCP     30s
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl logs nodejs-pod
Server running at port 3000
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ kubectl describe pod nodejs-pod
Name: nodejs-pod
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Wed, 10 Dec 2025 12:50:29 +0530
Labels: app=nodejs
Annotations: <none>
Status: Running
IP: 10.244.0.121
IPs:
  IP: 10.244.0.121
Containers:
  nodejs:
    Container ID: docker://34f81717339d6e06734e1fe656df9a8fb2579e063822a2f51ddc29e25d1b9866
    Image: node:18
    Image ID: docker-pullable://node@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
    Port: 3000/TCP
```

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1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg10
Environment:  <none>
Mounts:
  /usr/src/app from app-code (rw)
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-g9hg7 (ro)
Conditions:
  Type              Status
  PodReadyToStartContainers  True
  Initialized         True
  Ready               True
  ContainersReady      True
  PodScheduled        True
Volumes:
  app-code:
    Type:          ConfigMap (a volume populated by a ConfigMap)
    Name:          nodejs-app-config
    Optional:      false
  kube-api-access-g9hg7:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:          kube-root-ca.crt
    Optional:              false
    DownwardAPI:           true
QoS Class:               BestEffort
Node-Selectors:           <none>
Tolerations:              node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                          node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type      Reason      Age   From      Message
  ----      -
  Normal    SandboxChanged  8m35s  kubelet   Pod sandbox changed, it will be killed and re-created.
  Normal    Pulled        8m34s  kubelet   Container image "node:18" already present on machine
  Normal    Created       8m34s  kubelet   Created container: nodejs
  Normal    Started       8m34s  kubelet   Started container nodejs
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ minikube service nginx-service

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NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32401

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🔗 Opening service default/nginx-service in default browser...
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$ Gtk-Message: 21:05:17.927: Not loading module "atk-bridge": The functionality is provide
d by GTK natively. Please try to not load it.
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg10$
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